Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

HBASE PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Databases

TOTAL: 10 QUESTIONS

Q1 What type of database is HBase and what are its core strengths? — Threat Model

Q2 How does HBase guarantee consistency and handle transactions? — Architecture

Q3 What index types does HBase support and when do you use each? — Configuration

Q4 How do you design a schema or data model in HBase? — Hardening

Q5 How does HBase scale horizontally? — SSO / IdP

Q6 How do you monitor and tune slow queries in HBase? — Observability

Q7 What backup and restore strategies does HBase support? — Lifecycle

Q8 How do you handle schema migrations in HBase? — Compliance

Q9 What security features does HBase provide? — Testing

Q10 What are common anti-patterns when using HBase in production? — Tuning

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 HBase is a relational, document, key-value,

Mitigation

HBase is a relational, document, key-value, graph, or time-series store. Its strengths such as ACID guarantees, horizontal scale, or flexible schema guide the workloads it is best suited for.

A2 HBase provides either full ACID transactions

Primitives

HBase provides either full ACID transactions or eventual consistency depending on its CAP theorem positioning. Transaction support varies from none (simple K/V stores) to serializable isolation (relational DBs).

A3 HBase offers B-tree, hash, full-text, spatial,

Hardening

HBase offers B-tree, hash, full-text, spatial, or composite indexes depending on the engine. Choose based on query patterns: B-tree for range queries, hash for equality lookups, full-text for search.

A4 Schema design in HBase involves normalizing

Gotchas

Schema design in HBase involves normalizing relations (relational DB) or denormalizing for access patterns (document/key-value). Understanding query needs upfront prevents costly migrations later.

A5 HBase scales via replication (read replicas)

Federation

HBase scales via replication (read replicas) for read-heavy workloads and sharding (partitioning data by key) for write-heavy ones. Some HBase implementations handle this automatically; others require manual configuration.

A6 HBase provides a slow query log,

Observability

HBase provides a slow query log, explain/explain plan, or profiling tools to surface inefficient queries. Common fixes include adding indexes, rewriting queries, or increasing cache size.

A7 HBase supports logical backups (export SQL

Automation

HBase supports logical backups (export SQL or BSON), physical backups (filesystem snapshot), and continuous replication to a standby. Point-in-time recovery requires WAL/oplog archiving on top of backups.

A8 Schema migrations in HBase are managed

Governance

Schema migrations in HBase are managed with versioned migration files applied in order by a migration tool. Migrations should be idempotent and tested in staging before production to avoid downtime.

A9 HBase supports role-based access control,

Assessment

HBase supports role-based access control, encrypted connections (TLS), encryption at rest, and audit logging. Always restrict user privileges to the minimum needed and disable anonymous access in production.

A10 Common mistakes include selecting all

Optimization

Common mistakes include selecting all columns instead of needed fields, missing indexes on frequently queried columns, running migrations without backups, and storing large blobs directly in the database instead of object storage.